

DB Sodium Bicarbonate 8.4% w/v Injection

DESCRIPTION: Clear, colourless, odourless solution.

COMPOSITION: Each 10 ml contains:

Sodium Ion	10mEq
Bicarbonate Ion	10mEq

PHARMACODYNAMICS: Sodium bicarbonate in water dissociates to provide sodium and bicarbonate ions. Sodium is the principal cation of the extracellular fluid and plays a large part in the therapy of fluid and electrolyte disturbances. Bicarbonate is a normal constituent of body fluids and the normal plasma level ranges from 24 to 31 mmol/L.

PHARMACOKINETICS: Administration of sodium bicarbonate will increase the plasma bicarbonate concentration and help restore the plasma within the normal range (pH 7.37-7.42). Changes in acid-base balance also stimulate compensation ion-exchange mechanisms. When the extracellular hydrogen ion concentration increases, as in acidosis, there is a redistribution of potassium ions from intracellular to extracellular fluid. Administration of Sodium Bicarbonate can cause a redistribution of potassium ions into cells in patients with acidosis, by decreasing the plasma pH. The urinary pH will be increased by sodium bicarbonate in patients with normal renal function. Alkalinising the urine can increase the solubility of certain weak acids, and can increase the ionisation and urinary excretion of lipid-soluble organic acids (e.g. phenobarbitone, salicylates).

INDICATION: Sodium bicarbonate is used as an alkalinising agent in the treatment of metabolic acidosis which may occur in many conditions including diabetes, starvation, hepatitis, cardiac arrest, shock, severe dehydration, renal insufficiency, severe diarrhoea, Addison's disease or administration of acidifying salts (e.g. excessive sodium chloride, calcium chloride, ammonium chloride). Sodium bicarbonate is also used to increase urinary pH in order to increase the solubility of certain weak acids (e.g. cystine, sulphonamides, uric acid) and in the treatment of certain intoxications (e.g. methanol, phenobarbitone, salicylates) to decrease renal absorption of the drug or to correct acidosis.

RECOMMENDED DOSAGE: Sodium Bicarbonate Injection is administered by the intravenous route. Dosage of sodium bicarbonate injection is determined by the severity of the acidosis, appropriate laboratory determinations, and the patient's age, weight and clinical condition. In cardiac arrest, an initial dose of 1 mEq/kg may be given, followed by 0.5 mEq/kg at ten minute intervals. In infants (up to 2 years of age) a 4.2% solution is recommended for i.v. administration at a dose not to exceed 8 mmol/kg/day. Slow administration rates and a 4.2% solution are recommended in neonates to minimize the possibility of producing hypernatraemia, decreasing cerebrospinal fluid pressure and inducing intracranial haemorrhage. In less urgent forms of metabolic acidosis, sodium bicarbonate injection may be added to other i.v. fluids (See Incompatibilities). The amount of bicarbonate to be given to older children and adults over a 4 to 8 hour period is approximately 2 to 5 mmol/kg of bodyweight, depending upon the severity of the acidosis as judged by the lowering of the total CO₂ content, blood pH and clinical condition of the patient. Bicarbonate therapy should always be planned in a stepwise fashion since the degree of response from a given dose is not precisely predictable. In general, it is unwise to attempt full correction of a low total CO₂ content during the first 24 hours of therapy, since this may be accompanied by an unrecognized alkalosis because of a delay in the readjustment of ventilation to normal.

ROUTE OF ADMINISTRATION: i.v infusion

CONTRAINDICATIONS: Sodium bicarbonate is contraindicated in patients with renal failure, respiratory or metabolic alkalosis, hypoventilation, hypernatraemia, hypertension, oedema, congestive heart failure, a history of urinary calculi and consistent potassium depletion or hypocalcaemia. It is also generally contraindicated in patients with excessive chloride loss from vomiting or continuous GI suctioning, and in patients at risk of developing diuretic-induced hypochlorhaemic alkalosis.

WARNINGS & PRECAUTIONS: Laboratory determination of the patient's acid-base status is recommended before and during treatment to minimize the possibility of overdosage and resultant metabolic alkalosis. Accidental extravascular injection of hypertonic solutions may cause vascular irritation or sloughing. Rapid injection (10 mL/min) of hypertonic sodium bicarbonate injection solutions into neonates and children under 2 years of age may produce hypernatraemia, a decrease in cerebrospinal fluid pressure and possible intracranial haemorrhage. In emergency situations, such as cardiac arrest, the risk of rapid infusion of the drug must be weighed against the potential for death from acidosis. It should also be noted that administration of sodium bicarbonate to children undergoing cardiopulmonary resuscitation may worsen respiratory acidosis. To minimise the risks of pre-existing hypokalaemia and/or hypocalcaemia, these electrolyte disturbances should be corrected prior to initiation of, or concomitantly with, sodium bicarbonate therapy. Whenever respiratory acidosis is present with metabolic acidosis, both pulmonary ventilation and perfusion must be adequately supported to get rid of excess carbon dioxide. Sodium Bicarbonate 8.4% solution may be administered intravenously or following dilution to isotonicity (1.5%) subcutaneously. For subcutaneous administration, an isotonic solution (1.5%) of sodium bicarbonate may be prepared by diluting 1ml of 8.4% sodium bicarbonate with 4.6ml of sterile water for injections. Do not use the injection if it contains precipitate. Do not use unless the solution is clear and the container and seal are intact. Discard any unused portion.

Precautions:

- 1)History of Congestive Heart Failure
- 2)History of Renal Impairment
- 3)Cirrhosis
- 4)Hypertension
- 5)Concurrent corticosteroids

INTERACTION WITH OTHER MEDICAMENTS: Combination containing any of the following medication, depending on the amount present, may also interact with this medication. Adrenocorticoids, mineralocorticoids, corticotrophin, amphetamines, quinidine, androgens, steroids, anabolic, antimuscarinics, ciprofloxacin, diuretics, potassium-depleting (such as bumetanide, thiazide, ethacrynic acid, frusemide, indapamide), mecamlamine, methenamine, mexiletine. Potassium supplement, salicylates, benzylpenicillin, carmustine, glycoferronium bromide, isoprenaline hydrochloride and suxamethonium chloride.

USE IN PREGNANCY & LACTATION: The use of sodium bicarbonate injection, as with any drug, in pregnant women should only be undertaken if the expected benefit outweighs the possible risk to the mother and foetus or child.

SIDE EFFECTS: Common side effects following the administration of parenteral sodium bicarbonate are belching or swelling of stomach due to the release of carbon dioxide, stomach cramps and unusual increase in thirst. Alkalosis and/or hypokalaemia may ensue as a result of prolonged use or over correction of the bicarbonate deficit. Accidental extravasation of i.v. hypertonic solutions of Sodium Bicarbonate have been reported to cause chemical cellulitis, with tissue necrosis, ulceration or sloughing at the site of infiltration. Prompt elevation of the part, warmth and local injection of lignocaine or hyaluronidase are recommended to prevent sloughing of extravasated i.v. infusions. Hyperirritability or tetany may occur, caused by rapid shifts of free ionised calcium or due to serum protein alterations arising from the pH changes.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Symptoms: Metabolic alkalosis which may be accompanied compensatory hyperventilation, paradoxical acidosis of the cerebrospinal fluid, severe hypokalaemia, hyperirritability or tetany.

Treatment: The bicarbonate should be stopped and the patient managed according to the degree of alkalosis present. To control the symptoms of alkalosis the patient should rebreathe expired air. Sodium chloride injection 0.9% may be given intravenously, potassium chloride also may be indicated if there is hypokalaemia.

Calcium gluconate may be used to control hyperirritability and tetany which can occur in severe alkalosis. Ammonium chloride may also be indicated as an acidifying agent in severe cases.

INCOMPATIBILITIES: Sodium bicarbonate is incompatible with acids, acidic salts, many alkaloidal salt, aspirin, bismuth salicylate, isoproterenol hydrochloride, norepinephrine bitartrate, dopamine hydrochloride and tubocurarine chloride. Sodium bicarbonate solutions should not be mixed with calcium or magnesium salts, cisplatin, dobutamine hydrochloride, labetalol hydrochloride or oxytetracycline hydrochloride as this may result in formation of insoluble precipitates.

STORAGE CONDITIONS: Store below 30°C. Protect from light. Keep out of reach of children. *JAUHKAN DARIPADA KANAK-KANAK.*

PACK SIZE: Packs of 10 ampoules x 10ml, 30 ampoules x 10ml, 50 ampoules x 10ml and 100 ampoules x 10ml.

SHELF LIFE:

Please refer to outer package.

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD
Lot 2599, Jalan Seruling 59 Kawasan 3,
Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA.